Recombination without Exhaustion

The hand proof, aligned item by item with its Isabelle verification

11 November 2026

Background. Shortly after Jacob Lettie found his refutation of Purity of Pure at type $t \to t$ (see his hand proof in this folder), a strengthening was found that drops Exhaustion. Lettie's ten premises include zeroary Exhaustion at two propositions and the existence of a universal proposition. Here Logical Purity, Application and PP_u, all ordinary premises over H + PE, refute the existence of a universal proposition outright (Theorem 1). Since a fundamental proposition is universal by unary Recombination, PP_u is refuted by Logical Purity, Application, unary Recombination and a fundamental proposition, with no Exhaustion or Persistence (the Corollary). This is the Recombination-only reading of Goodman's question.

Status. Each result below carries, in brackets, the name of its theorem in the Isabelle session `Goodman_Integration_Recombination_Only` or in Lettie's kernel on which it builds. The sub-steps (b1)–(b5), (d5) and (i)–(vii) of (2.7) are intermediate facts inside the proof of the cited theorem. The session builds under Isabelle2026 on Lettie's unmodified kernel `PoP_Main`, with no oracles; see Appendix A. The proof has been checked in Isabelle and against the theory files, but it has not yet had an independent human review.

1 Setting

Types and notation. Propositions have type t. Put $u := t \to t$ and $c := u \to t$, and write $\pi := \mathrm{Pure}_u$, a constant of type c (`rc_p`). Letters p, q, x, y, z, r, b range over type t, and S, X, Y over type u. Put

$$\top := \forall p(p \to p), \qquad \Box A := (A = \top), \qquad \Diamond A := {\sim}\Box{\sim}A.$$

Base logic. The base logic is H + PE, formalized as Lettie's calculus `U_proves`. Its theorems are those of H (classical simple type theory with $\beta\eta$-conversion, Leibniz identity, propositional tautologies, universal instantiation and existential generalization), closed under four rules: Propositional Equivalence (from a theorem $A \leftrightarrow B$ at type t, infer $A = B$), modus ponens, Generalization, and Instantiation ($\exists$-elimination). We use the following derived facts:

Necessitation: from $\vdash A$ infer $\vdash \Box A$	`U_nec`
K-rule: from $\vdash A \to B$ infer $\vdash \Box A \to \Box B$	`u_rc_K`
T: $\Box A \to A$; 4: $\Box A \to \Box\Box A$	`U_modal_T`, `U_modal_4`
$\Box(A \leftrightarrow B) \to A = B$	`u_rc_box_iff_eq`
Leibniz: $a = b \to (Fa \to Fb)$; congruence for $=$ at type t	`u_rc_LL`, `u_rc_congr`
$\beta\eta$-convertible formulas are equivalent	`U_beta_eta_equiv`

Below, "by β" cites the last fact.

Ordinary premises. The calculus has no rule for premises. An added principle can therefore occur only as the antecedent of an implication that is a theorem of the base logic. Propositional Equivalence and Necessitation then apply only to such premise-free theorems, never to a premise. "Under premises Π, A" means that $\bigwedge \Pi \to A$ is a theorem of H + PE.

The added principles. *Logical Purity* is the schema $\mathrm{Pure}_\sigma(M)$ for closed terms $M : \sigma$ that contain no constants (`pp_purity_schema`). Since Pure is itself a constant (`pp_Pure`), Logical Purity says nothing about π; this is immediate from the definitions. *Application* at (σ, τ) is

$$\mathrm{App}_{\sigma,\tau} := \forall X^{\sigma\to\tau} \forall y^{\sigma} \big((\mathrm{Pure}_{\sigma\to\tau}(X) \,\&\, \mathrm{Pure}_\sigma(y)) \to \mathrm{Pure}_\tau(Xy)\big).$$

This is `pp_application_closure` $\sigma\ \tau$, used through `u_rc_ac`. The target is $\mathrm{PP}_u := \mathrm{Pure}_c(\pi)$ (`pp_target_PP`).

The proof uses exactly twenty instances of Logical Purity and Application, together with PP_u. They are listed in Table 1 with the steps that use them. The closed terms named there are defined in Sections 3–6.

2 The results

Following Lettie, put (`rcR`, `rcUni`, `rcExUni`)

$$R(x,y) := \forall Y^{u}\big(\pi(Y) \to (Yy \to \Diamond Yx)\big), \qquad \mathrm{Uni}(x) := \forall y\, R(x,y).$$

A proposition x is *universal* if $\mathrm{Uni}(x)$.

Theorem 1. $\mathrm{H} + \mathrm{PE} \vdash \mathrm{LA} \to \big(\mathrm{PP}_u \to {\sim}\exists r\, \mathrm{Uni}(r)\big)$. [`ro_thm1_U`]

In the repository's ordinary-premise relation $\vdash_{\mathrm{CEV}_s}$ over C, the premise set consisting of PP_u and the full Logical Purity and Application schemas derives ${\sim}\exists r\, \mathrm{Uni}(r)$. [`ro_thm1_CEVs`]

Premise	Instance	Used in
L1	$\mathrm{Pure}_t(\top)$	(Cl1) for G
L2	$\mathrm{Pure}(\mathsf{and})$, $\mathsf{and} := \lambda p \lambda q.(p \,\&\, q)$	(Cl2) for G; $\mathrm{Pure}_t(a)$
L3	$\mathrm{Pure}(m)$, $m := \lambda S. \forall p (Sp \to p)$	(Cl5) for G; $\mathrm{Pure}_t(a)$
L4	$\mathrm{Pure}(\widehat{W})$	$\pi(W)$
L5	$\mathrm{Pure}(\hat{\Phi})$	$\mathrm{Pure}_u(\Phi)$
L6, L7	$\mathrm{Pure}(\hat{P})$, $\mathrm{Pure}(\hat{P}')$	Φ preserves pure truths
L8	$\mathrm{Pure}(\hat{Q})$	Q_a pure (Step 6.2)
L9	$\mathrm{Pure}(Z)$, $Z := \lambda a \lambda Y \lambda x.(a \,\&\, Yx)$	Step 6.1
L10	$\mathrm{Pure}(\mathsf{comp})$, $\mathsf{comp} := \lambda Y \lambda S \lambda x. Y(Sx)$	Step 6.2
A1	$\mathrm{App}_{t,t\to t}$	(Cl2) for G; $\mathrm{Pure}_t(a)$
A2	$\mathrm{App}_{t,t}$	(Cl2), (Cl3) for G; $\mathrm{Pure}_t(a)$
A3	$\mathrm{App}_{u,t}$	(Cl4), (Cl5) for G; $\mathrm{Pure}_t(a)$
A4, A5	$\mathrm{App}_{c,u\to u}$, $\mathrm{App}_{u,u}$	$\pi(W)$; A5 also in Steps 6.1, 6.2
A6	$\mathrm{App}_{c,u}$	$\mathrm{Pure}_u(\Phi)$
A7, A8	$\mathrm{App}_{c,t\to u}$, $\mathrm{App}_{t,u}$	Φ preserves pure truths; Step 6.2
A9	$\mathrm{App}_{t,u\to u}$	Step 6.1
A10	$\mathrm{App}_{u,u\to u}$	Step 6.2
PP_u	$\mathrm{Pure}_c(\pi)$	(Cl4); $\pi(W)$; $\mathrm{Pure}_t(a)$; $\mathrm{Pure}_u(\Phi)$; κ; Step 6.2

Table 1: The premises. L1–L4 with A1–A5 form `roHC0`; L5–L10 with A6–A10 form `roHF`; their conjunction LA is `roLA`.

Corollary. *With* $\mathrm{EF} := \exists r\, \mathrm{Fun}_t(r)$ *and Lettie's* H5 *(Section 7),* $\mathrm{H} + \mathrm{PE} \vdash \mathrm{LA} \to (\mathrm{EF} \to (\mathrm{H5} \to {\sim}\mathrm{PP}_u))$. [`ro_cor_EF_U`]

In $\vdash_{\mathrm{CEV}_s}$*, the repository's Recombination package for the question derives* ${\sim}\mathrm{PP}_u$*.* [`ro_recombination_refutes_PP`]

No Exhaustion, Persistence, or Purity of Fun is used.

3 The closure lemma

Fix a closed term $\Theta : u$. It may contain constants. Section 5 takes $\Theta := \Phi$.

Definitions. The closure condition with its two parameters abstracted is

$$\widehat{\mathrm{Cl}} := \lambda c' \lambda T \lambda X.\ (\mathrm{Cl1} \,\&\, \mathrm{Cl2} \,\&\, \mathrm{Cl3} \,\&\, \mathrm{Cl4} \,\&\, \mathrm{Cl5}) \qquad (\texttt{ro_Cl}),$$

where, writing π for c' and Θ for T,

(Cl1) $X\top$;

(Cl2) $\forall p\,\forall q\big((Xp \,\&\, Xq) \to X(p \,\&\, q)\big)$;

(Cl3) $\forall p\big(Xp \to X(\Theta p)\big)$;

(Cl4) $\forall S\big(\pi(S) \to X(\pi(S))\big)$;

(Cl5) $\forall S\big((\pi(S) \,\&\, \forall p(Sp \to Xp)) \to X(m(S))\big)$.

Put

$$\begin{aligned}
\mathrm{Cl}(X) &:= \widehat{\mathrm{Cl}}\,\pi\,\Theta\,X && (\texttt{roCl}),\\
\widehat{W} &:= \lambda c'\lambda T\lambda q.\,\forall X(\widehat{\mathrm{Cl}}\,c'\,T\,X \to Xq) && (\texttt{ro_W}),\\
W &:= \widehat{W}\,\pi\,\Theta && (\texttt{roW}),\\
a &:= m(W) \,\&\, \pi(W) && (\texttt{roa}),\\
G &:= \lambda p.(\mathrm{Pure}_t(p) \,\&\, p) && (\texttt{ro_G}),\\
\mathrm{Pres}(\Theta) &:= \forall p\big((\mathrm{Pure}_t(p) \,\&\, p) \to \Theta p\big) && (\texttt{roPres}).
\end{aligned}$$

The terms and, m, $\widehat{\mathrm{Cl}}$, $\widehat{W}$ are closed and contain no constants (`ro_closed`, `ro_consts`). By β, $\mathrm{Cl}(X)$ is the conjunction (Cl1)–(Cl5) (`ro_Cl_iff`), and

$$Wq \leftrightarrow \forall X(\mathrm{Cl}(X) \to Xq) \quad (\texttt{ro_W_iff}), \qquad m(S) \leftrightarrow \forall p(Sp \to p) \quad (\texttt{ro_m_iff}).$$

We use W in two ways:

Elimination: $Wq \to (\mathrm{Cl}(X) \to Xq)$. [`ro_W_elim`]

Introduction: from $P \to (\mathrm{Cl}(X) \to Xq)$, with X fresh, infer $P \to Wq$. [`ro_W_intro`]

The predicate G contains the constant Pure_t. It is only used to instantiate $\forall X$, and its purity is never asserted.

Lemma 1 (Closure)**.** *The following hold.*

(a) $\vdash \mathrm{Cl}(W)$. [`ro_ClW`]

(b) $\vdash a \to \Theta a$. [`ro_base`]

(c) $\vdash \Box(a \to \Theta a)$. [`ro_boxed`]

(d) Under L1–L4, A1–A5, and PP_u*:* $\mathrm{Pure}_u(\Theta) \to \big(\mathrm{Pres}(\Theta) \to (\mathrm{Pure}_t(a) \,\&\, a)\big)$. [`ro_closure_true`]

Parts (a)–(c) use no added principle.

Proof. *(a)* Each clause is proved by introduction for W, from the same clause for an arbitrary X with $\mathrm{Cl}(X)$.

- (Cl1): $\mathrm{Cl}(X) \to X\top$, so $W\top$. [ro_ClW_1]
- (Cl2): elimination at p and at q gives Xp and Xq from $Wp \,\&\, Wq$ and $\mathrm{Cl}(X)$. Then (Cl2) for X gives $X(p \,\&\, q)$. Generalize p, q. [ro_ClW_2]
- (Cl3): from Wp and $\mathrm{Cl}(X)$, elimination gives Xp, and (Cl3) for X gives $X(\Theta p)$. [ro_ClW_3]
- (Cl4): from $\pi(S)$ and $\mathrm{Cl}(X)$, (Cl4) for X gives $X(\pi(S))$. [ro_ClW_4]
- (Cl5): assume $\pi(S) \,\&\, \forall p(Sp \to Wp)$ and $\mathrm{Cl}(X)$. For each p, $Sp \to Wp$ and elimination give $Sp \to Xp$. By Generalization, $\forall p(Sp \to Xp)$, so (Cl5) for X gives $X(m(S))$. [ro_ClW_5]

(b) Instantiate the clauses of (a) at W itself.

(b1) (Cl4) at $S := W$: $\pi(W) \to W(\pi(W))$.

(b2) (Cl5) at $S := W$, with the theorem $\forall p(Wp \to Wp)$: $\pi(W) \to W(m(W))$.

(b3) (Cl2) at $m(W), \pi(W)$: $(W(m(W)) \,\&\, W(\pi(W))) \to Wa$.

(b4) (Cl3) at a: $Wa \to W(\Theta a)$.

(b5) $m(W) \leftrightarrow \forall p(Wp \to p)$, instantiated at Θa: $m(W) \to (W(\Theta a) \to \Theta a)$.

Assume a, that is, $m(W)$ and $\pi(W)$. Steps (b1)–(b3) give Wa, (b4) gives $W(\Theta a)$, and (b5) gives Θa.

(c) Necessitation applied to (b), a theorem without premises.

(d) Assume the premises, $\mathrm{Pure}_u(\Theta)$ and $\mathrm{Pres}(\Theta)$.

(d1) $\mathrm{Cl}(G)$, clause by clause, using $Gp \leftrightarrow (\mathrm{Pure}_t(p) \,\&\, p)$ by β (ro_G_iff). [ro_ClG]

- (Cl1): $\top$ is true and pure by L1. [ro_ClG_1]
- (Cl2): let Gp and Gq. Then $p \,\&\, q$ is true. By L2, A1 and A2, $\mathrm{Pure}_t(\mathsf{and}\, p\, q)$. By β, $\mathrm{Pure}_t(p \,\&\, q)$ (ro_pure_be). [ro_ClG_2]
- (Cl3): let Gp. $\mathrm{Pres}(\Theta)$ gives Θp. $\mathrm{Pure}_u(\Theta)$, $\mathrm{Pure}_t(p)$ and A2 give $\mathrm{Pure}_t(\Theta p)$. [ro_ClG_3]

- (Cl4): let $\pi(S)$, which is $\mathrm{Pure}_u(S)$. PP_u and A3 give $\mathrm{Pure}_t(\pi(S))$, and $\pi(S)$ is true. [ro_ClG_4]
- (Cl5): let $\pi(S)$ and $\forall p(Sp \to Gp)$. Since $Gp \to p$, $\forall p(Sp \to p)$, which is $m(S)$ by β. L3, $\mathrm{Pure}_u(S)$ and A3 give $\mathrm{Pure}_t(m(S))$. [ro_ClG_5]

(d2) $m(W)$: elimination at $X := G$ and (d1) give $Wp \to Gp$, hence $Wp \to p$. Generalize p, and apply β. [ro_mW]

(d3) $\pi(W)$: PP_u, L4 and A4 give $\mathrm{Pure}_{u\to u}(\widehat{W}\pi)$. Then $\mathrm{Pure}_u(\Theta)$ and A5 give $\mathrm{Pure}_u(W)$, which is $\pi(W)$. [ro_piW]

(d4) $\mathrm{Pure}_t(a)$: L3, (d3) and A3 give $\mathrm{Pure}_t(m(W))$. PP_u, (d3) and A3 give $\mathrm{Pure}_t(\pi(W))$. Then L2, A1, A2 and β give $\mathrm{Pure}_t(m(W)\,\&\,\pi(W))$. [ro_pure_a]

(d5) By (d2) and (d3), a is true. With (d4), this proves (d).

■

The key clause is (Cl4). At $S = W$ it places the guard $\pi(W)$ in W, so the conditional $a \to \Theta a$ holds in the base logic. Clause (Cl1) is not used in (b); it is the clause that accounts for premise L1.

4 The relativized Lettie construction

Definitions. The following terms are closed and contain no constants; in Isabelle these facts are `ro_rel_closed` and `ro_rel_consts`.

$$\hat{R} := \lambda c'\lambda b\lambda x\lambda y.\,\forall Y(c'(Y) \to (Yy \to \Diamond(b\,\&\,Yx))) \qquad \text{(ro_R)},$$
$$\hat{F} := \lambda c'\lambda b\lambda z.\,\Box(b \to (\sim\hat{R}c'bz(\sim z)\,\&\,\sim\hat{R}c'b(\sim z)z)) \qquad \text{(ro_F)},$$
$$\hat{Q} := \lambda c'\lambda b\lambda z.(\hat{F}c'bz \to z) \qquad \text{(ro_Q)},$$
$$\hat{P} := \lambda c'\lambda b\lambda z.(\sim z\,\&\,\sim\hat{F}c'bz) \qquad \text{(ro_P)},$$
$$\hat{P}' := \lambda c'\lambda b\lambda z.(z\,\&\,\sim\hat{F}c'bz) \qquad \text{(ro_Pp)},$$
$$\hat{\Phi} := \lambda c'\lambda b.(b\,\&\,(c'(\hat{P}c'b)\,\&\,c'(\hat{P}'c'b))) \qquad \text{(ro_Phi)}.$$

For a proposition b, let $R_b(x,y) := \hat{R}\pi bxy$, $F_b(z) := \hat{F}\pi bz$, $Q_b := \hat{Q}\pi b$, $P_b := \hat{P}\pi b$, $P'_b := \hat{P}'\pi b$, and $\Phi := \hat{\Phi}\pi$ (roR, roF, roQf, roP, roPp, roPhi). Put

$$C_b(z) := \sim R_b(z,\sim z)\,\&\,\sim R_b(\sim z,z) \quad \text{(roC)}, \qquad \kappa(b) := \pi(P_b)\,\&\,\pi(P'_b) \quad \text{(rok)}.$$

By β (`roR_iff`, `roF_iff`, `roQ_iff`, `roP_iff`, `roPp_iff`, `roPhi_iff`):

$$R_b(x,y) \leftrightarrow \forall Y\big(\pi(Y) \to (Yy \to \Diamond(b \,\&\, Yx))\big), \qquad F_b(z) \leftrightarrow \Box(b \to C_b(z)),$$
$$Q_b(z) \leftrightarrow (F_b(z) \to z), \qquad P_b(z) \leftrightarrow (\sim z \,\&\, \sim F_b(z)),$$
$$P'_b(z) \leftrightarrow (z \,\&\, \sim F_b(z)), \qquad \Phi(b) \leftrightarrow (b \,\&\, \kappa(b)).$$

The rule "from $P \to (\pi(Y) \to (Yy \to \Diamond(b \,\&\, Yx)))$ with Y fresh, infer $P \to R_b(x,y)$" is `roR_intro`. Elimination, the instance of $R_b(x,y)$ at a given Y, is `roR_elim`.

Every result in this section is a premise-free theorem, for arbitrary proposition terms b, x and z.

Lemma 2 (Relative lemma)**.** $\vdash \Box(b \to \kappa(b)) \to F_b(Q_b(x))$. [`ro_relative`]

The proof replays the logical part of Lettie's argument with the guard b. It passes through seven premise-free theorems.

(2.1) $F_b(x) \to \Box F_b(x)$. [`ro_lemA`]

(2.2) $F_b(x) \to Q_b(x) = x$. [`ro_lemB`]

(2.3) $F_b(\sim x) \leftrightarrow F_b(x)$. [`ro_lemC`]

(2.4) $P'_b(z) = P_b(\sim z)$. [`ro_ident`]

(2.5) $\sim Q_b(x) \to F_b(Q_b(x))$. [`ro_lemD`]

(2.6) $\Box \sim P_b(Q_b(x))$. [`ro_cor`]

(2.7) $(b \,\&\, \kappa(b)) \to C_b(Q_b(x))$. [`ro_guarded`]

Proof of (2.1)–(2.7). (2.1) By β, $F_b(x) \leftrightarrow \Box B$ with $B := b \to C_b(x)$. Axiom 4 gives $\Box B \to \Box\Box B$. The K-rule applied to $\Box B \to F_b(x)$ gives $\Box\Box B \to \Box F_b(x)$.

(2.2) By β, $Q_b(x) \leftrightarrow (F_b(x) \to x)$. Hence $\vdash F_b(x) \to (Q_b(x) \leftrightarrow x)$, and the K-rule gives $\Box F_b(x) \to \Box(Q_b(x) \leftrightarrow x)$. Combine with (2.1) and $\Box(A \leftrightarrow B) \to A = B$.

(2.3) Propositional Equivalence gives $\sim\sim x = x$. So, by congruence, $R_b(\sim x, \sim\sim x) \leftrightarrow R_b(\sim x, x)$ and $R_b(\sim\sim x, \sim x) \leftrightarrow R_b(x, \sim x)$. So $C_b(\sim x) \leftrightarrow C_b(x)$, with the two conjuncts exchanged, and Propositional Equivalence gives $C_b(\sim x) = C_b(x)$. Congruence in $\Box(b \to \cdot)$ and β give the claim.

(2.4) By β, $P'_b(z) \leftrightarrow (z \,\&\, \sim F_b(z))$ and $P_b(\sim z) \leftrightarrow (\sim\sim z \,\&\, \sim F_b(\sim z))$. By (2.3) the right-hand sides are equivalent. Propositional Equivalence gives the identity.

(2.5) By β, $\sim Q_b(x) \to F_b(x)$. By (2.2) and symmetry, $F_b(x) \to x = Q_b(x)$. Leibniz's law for F_b gives $F_b(x) \to F_b(Q_b(x))$.

(2.6) Let $q := Q_b(x)$. By β, $P_b(q) \leftrightarrow (\sim q \,\&\, \sim F_b(q))$. By (2.5), $\sim q \to F_b(q)$, so $\vdash \sim P_b(q)$. This is a premise-free theorem, so Necessitation applies.

(2.7) Let $q := Q_b(x)$. We first collect seven theorems.

(i) $(b \,\&\, \sim C_b(q)) \to (q \,\&\, \sim F_b(q))$. If $F_b(q)$, then $\Box(b \to C_b(q))$ by β, and T gives $b \to C_b(q)$, a contradiction. So $\sim F_b(q)$, and the contrapositive of (2.5) gives q.

(ii) $(q \,\&\, \sim F_b(q)) \to P_b(\sim q)$. By β, $P_b(\sim q) \leftrightarrow (\sim\sim q \,\&\, \sim F_b(\sim q))$. Use (2.3) at q.

(iii) $(q \,\&\, \sim F_b(q)) \to P'_b(q)$, by β.

(iv) $\Box\sim(b \,\&\, P_b(q))$. Apply the K-rule to the tautology $\sim P_b(q) \to \sim(b \,\&\, P_b(q))$, then use (2.6).

(v) $\Diamond(b \,\&\, P'_b(\sim q)) \leftrightarrow \Diamond(b \,\&\, P_b(q))$. By (2.4) at $\sim q$, $P'_b(\sim q) = P_b(\sim\sim q)$, and congruence turns this identity into $P'_b(\sim q) \leftrightarrow P_b(\sim\sim q)$. Also $\sim\sim q = q$, so congruence gives $P_b(\sim\sim q) \leftrightarrow P_b(q)$. Chaining the two biconditionals, Propositional Equivalence gives $P'_b(\sim q) = P_b(q)$, and congruence in $\Diamond(b \,\&\, \cdot)$ gives (v).

(vi) $R_b(q, \sim q) \to (\pi(P_b) \to (P_b(\sim q) \to \Diamond(b \,\&\, P_b(q))))$, the instance at $Y := P_b$.

(vii) $R_b(\sim q, q) \to (\pi(P'_b) \to (P'_b(q) \to \Diamond(b \,\&\, P'_b(\sim q))))$, the instance at $Y := P'_b$.

Now assume b, $\pi(P_b)$, $\pi(P'_b)$ and $\sim C_b(q)$, so $R_b(q, \sim q) \vee R_b(\sim q, q)$. By (i)–(iii), $P_b(\sim q)$ and $P'_b(q)$. In the first case (vi) gives $\Diamond(b \,\&\, P_b(q))$. In the second, (vii) and (v) give the same conclusion. Either contradicts (iv). The combination is propositional. ∎

Proof of Lemma 2. By (2.7), $\vdash (b \to \kappa(b)) \to (b \to C_b(Q_b(x)))$. The K-rule gives $\Box(b \to \kappa(b)) \to \Box(b \to C_b(Q_b(x)))$. The consequent is $F_b(Q_b(x))$ by β. ∎

The lemma is stated for each term x. Generalization would give $\forall x$, but only the instance $x := r$ is used below.

5 The required pure truth

Now take $\Theta := \Phi = \hat{\Phi}\pi$, which is closed, and let a and W be as in Section 3 for this Θ. Write Prem := LA & PP_u (`roPrem`, up to the order and grouping of conjuncts).

(5.1) Under Prem, $\mathrm{Pure}_u(\Phi)$: use PP_u, L5 and A6. [`ro_Phi_pure`]

(5.2) Under Prem, $\mathrm{Pres}(\Phi)$. Let b be pure and true. PP_u, L6, A7, $\mathrm{Pure}_t(b)$ and A8 give $\mathrm{Pure}_u(P_b)$; likewise L7 gives $\mathrm{Pure}_u(P'_b)$. These are $\pi(P_b)$ and $\pi(P'_b)$, so $\kappa(b)$ holds. With b and β, $\Phi(b)$ holds. [ro_Phi_pres]

(5.3) Under Prem, $\mathrm{Pure}_t(a)\,\&\,a$: Lemma 1(d) with (5.1) and (5.2). [ro_actual]

(5.4) $\vdash \Box(a \to \kappa(a))$, with no premises. By β, $\vdash (a \to \Phi(a)) \to (a \to \kappa(a))$. Apply the K-rule and use Lemma 1(c). [ro_certificate]

(5.5) $\vdash F_a(Q_a(x))$ for every x, with no premises: Lemma 2 at $b := a$, and (5.4). [ro_Fa]

The premises enter only at (5.1)–(5.3). Steps (5.4) and (5.5) are theorems of the base logic. No purity or truth assertion is necessitated.

6 The contradiction

Let r be a proposition term. With the closed terms $Z := \lambda a \lambda Y \lambda x.(a\,\&\,Yx)$ and $\mathsf{comp} := \lambda Y \lambda S \lambda x.Y(Sx)$ (ro_Z, ro_comp), β gives $Z\,a\,Y\,x \leftrightarrow (a\,\&\,Yx)$ and $\mathsf{comp}\,Y\,S\,x \leftrightarrow Y(Sx)$ (roZ_iff, roComp_iff).

(6.1) Under Prem and $\mathrm{Uni}(r)$, $R_a(r, y)$ for every y. [ro_step1, ro_uni_rel]

Let Y be pure and Yy, and put $Z_Y := Z\,a\,Y$.

- Z_Y is pure: L9, $\mathrm{Pure}_t(a)$ from (5.3) and A9, then $\mathrm{Pure}_u(Y)$ and A5.
- $Z_Y(y)$: by β, from the truth of a (5.3) and Yy.
- $\mathrm{Uni}(r)$ gives $R(r, y)$ (u_rcUni_inst); its instance at Z_Y (u_rcR_elim) gives $\Diamond Z_Y(r)$.
- By β, $\sim(a\,\&\,Yr) \to \sim Z_Y(r)$. The K-rule turns $\Diamond Z_Y(r)$ into $\Diamond(a\,\&\,Yr)$.

Introduction for R_a over Y gives $R_a(r, y)$.

(6.2) Under Prem and $\mathrm{Uni}(r)$, $R_a(e, \sim e)$, where $e := Q_a(r)$. [ro_step2, ro_Rae]

Let Y be pure and $Y(\sim e)$, and put $Y' := \mathsf{comp}\,Y\,Q_a$.

- $Q_a = \hat{Q}\pi a$ is pure: PP_u, L8, A7, $\mathrm{Pure}_t(a)$ and A8. Then Y' is pure: L10, $\mathrm{Pure}_u(Y)$ and A10, then $\mathrm{Pure}_u(Q_a)$ and A5.
- $Y'(\sim e)$. By (5.5) at $x := r$, $F_a(e)$. By (2.3), $F_a(\sim e)$. By (2.2) at $\sim e$, $Q_a(\sim e) = \sim e$. By symmetry and Leibniz's law, $Y(\sim e) \to Y(Q_a(\sim e))$, and β gives $Y'(\sim e)$.

- (6.1) at $y := {\sim}e$ gives $R_a(r, {\sim}e)$. Its instance at Y' gives $\Diamond(a \,\&\, Y'(r))$.
- By β, $Y'(r) \leftrightarrow Y(e)$. The K-rule gives $\Diamond(a \,\&\, Y(e))$.

Introduction for R_a over Y gives $R_a(e, {\sim}e)$.

(6.3) Under Prem, ${\sim}\mathrm{Uni}(r)$. [ro_contra]

By (5.5) and β, $\Box(a \to C_a(e))$. T gives $a \to C_a(e)$, and a is true by (5.3). So $C_a(e)$, whose first conjunct is ${\sim}R_a(e, {\sim}e)$. This contradicts (6.2).

(6.4) $\vdash$ Prem $\to {\sim}\exists r\, \mathrm{Uni}(r)$. Apply (6.3) to a variable r, then the Instantiation rule. [ro_thm1]

Proof of Theorem 1. Reorder the antecedent of (6.4) to separate PP_u (ro_thm1_U). For the CEV form, each conjunct of LA belongs to the Logical Purity or the Application schema (ro_LA_derivable), and PP_u is a member of the premise set (an Assumption step). The theorem transfers from U to C by U_sub_CEV, and modus ponens with the premises gives the conclusion (ro_thm1_CEVs). ■

7 Fundamental propositions and the corollary

Unary Recombination (pp_unary_recombination) is

$$\mathrm{Rec}_1 := \forall X^u \forall r\big((\mathrm{Pure}_u(X) \,\&\, \mathrm{Fun}_t(r)) \to (\Box Xr \to \forall z\, Xz)\big).$$

Lettie's H5 (rcH5) is $\mathrm{Pure}_{u\to u}(\mathsf{neg}) \,\&\, \mathrm{App}_{u,u} \,\&\, \mathrm{Rec}_1$, where $\mathsf{neg} := \lambda Y \lambda x.{\sim}Yx$ (k_C).

Proposition (Lettie)**.** $\vdash$ EF $\to$ (H5 $\to \exists r\, \mathrm{Uni}(r)$). [u_fun_ExUni]

Proof. This lemma is imported from Lettie's kernel and was checked there. Let $\mathrm{Fun}_t(r)$, and let Y be pure with Yy (u_fun_step). The operator $X := \mathsf{neg}\, Y$ is pure, since Pure(neg), $\mathrm{Pure}_u(Y)$ and $\mathrm{App}_{u,u}$ hold. By β, ${\sim}Yr \to Xr$, and the K-rule gives $\Box{\sim}Yr \to \Box Xr$. If $\Box{\sim}Yr$, then Rec_1 gives $\forall z\, Xz$, hence Xy, that is, ${\sim}Yy$: a contradiction. So $\Diamond Yr$. Introduction for R and Generalization give $\mathrm{Uni}(r)$ (u_uni_ctx), and existential generalization with Instantiation over EF give the claim. ■

Proof of the Corollary. Combine Theorem 1 with the Proposition (ro_cor_EF_U). Unique fundamentality implies EF (lo_EF_of_UF), which gives ro_cor_U. For the repository package, the members used are the Logical Purity schema (L1–L10 and neg), the Application schema, unique fundamentality, and Rec_1 (ro_cor_set). The package's other members,

"no other fundamentals" and zeroary Recombination, are not used. Adding PP_u to the package derives $\bot$ (`ro_recombination_PP_inconsistent`). ∎

Appendix A: verification record

Session `Goodman_Integration_Recombination_Only` = `PoP_Main` + `Goodman_RecOnly_Closure`, `Goodman_RecOnly_Relative`, `Goodman_RecOnly_Final`, `Goodman_RecOnly_Audit`, in `theories/recombination_only/` of the Goodman application. It is built by the application's checker, `./check_isabelle.sh -export`, serially and with a 60-second session timeout, under Isabelle2026. Lettie's files and the Bacon–Dorr core are unchanged. The audit theory checks 31 endpoint theorems for oracles (none), residual hypotheses (none), and flex-flex pairs (none). Its catalog and statement export are in `verification/audits/Goodman_Integration_Recombination_Only.Goodman_RecOnly_Audit/`. No proof uses `sorry`, `oops`, `quick_and_dirty`, or a new axiomatization. No proof uses the axiom-extension calculi `CEV_axiom_proves` or `goodman_book_proves`.

This note	Isabelle theorem or definition
$m, \widehat{\mathrm{Cl}}, \mathrm{Cl}, \widehat{W}, W, a, G, \mathrm{Pres}$	`ro_m`, `ro_Cl`, `roCl`, `ro_W`, `roW`, `roa`, `ro_G`, `roPres`
Premises L1–L4, A1–A5, PP_u	`roHC` (`roHC0` without PP_u)
Lemma 1(a), clause by clause	`ro_ClW_1`–`ro_ClW_5`, `ro_ClW`
Lemma 1(b), (c)	`ro_base`, `ro_boxed`
(d1), clause by clause	`ro_ClG_1`–`ro_ClG_5`, `ro_ClG`
(d2), (d3), (d4), (d)	`ro_mW`, `ro_piW`, `ro_pure_a`, `ro_closure_true`
$\hat{R}, \hat{F}, \hat{Q}, \hat{P}, \hat{P}', \hat{\Phi}$	`ro_R`, `ro_F`, `ro_Q`, `ro_P`, `ro_Pp`, `ro_Phi`
$R_b, F_b, Q_b, Q_b(z), P_b, P_b', C_b, \kappa, \Phi$	`roR`, `roF`, `roQf`, `roQ`, `roP`, `roPp`, `roC`, `rok`, `roPhi`
(2.1)–(2.7)	`ro_lemA`, `ro_lemB`, `ro_lemC`, `ro_ident`, `ro_lemD`, `ro_cor`, `ro_guarded`
Lemma 2	`ro_relative`
Premises L5–L10, A6–A10; Prem	`roHF`; `roPrem`
(5.1)–(5.5)	`ro_Phi_pure`, `ro_Phi_pres`, `ro_actual`, `ro_certificate`, `ro_Fa`
Z, comp	`ro_Z`, `ro_comp`
(6.1)–(6.4)	`ro_step1`, `ro_uni_rel`, `ro_step2`, `ro_Rae`, `ro_contra`, `ro_thm1`
Theorem 1	`ro_thm1_U`, `ro_thm1_CEVs`
Proposition (Lettie)	`u_fun_step`, `u_uni_ctx`, `u_fun_ExUni` (imported)
Corollary	`ro_cor_EF_U`, `ro_cor_U`, `ro_cor_set`, `ro_recombination_refutes_PP`, `ro_recombination_PP_inconsistent`